

1 Baseline Covariate Adjustment in Randomized Clinical Trials:
2 Objective Criteria for When and How to Adjust

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36 1 Abstract

37 **Background.** Covariate adjustment in randomized clinical trials remains methodologically
38 contested in applied practice despite a mature theoretical literature, mature regulatory guid-
39 ance from FDA, EMA, and ICH E9, and consistent simulation evidence. The disconnect
40 between literature and practice is partly attributable to the absence of objective criteria that
41 an investigator can apply without re-deriving the relevant theory; this paper develops such
42 criteria.

43 **Methods.** We synthesize the statistical literature on covariate adjustment in RCTs, the rele-
44 vant regulatory guidance, and illustrative simulation evidence on adjusted- versus-unadjusted
45 analyses across varying covariate-outcome correlations and sample sizes. Performance mea-
46 sures include bias, coverage, and efficiency gain (relative variance reduction), reported with
47 Monte Carlo standard errors following the ADEMP framework of¹. The synthesis distills into
48 objective criteria along three axes: when to adjust, which covariates to adjust for, and how
49 to specify the adjustment.

50 **Results.** The criteria converge: adjustment should always be applied when prognostic co-
51 variates are pre-specified; stratification variables used in the randomization must appear as
52 covariates in the analysis; the baseline measurement of the outcome variable should always
53 be included when available. Selection of additional covariates should rest on prior evidence
54 of prognostic value, with the number of covariates kept small relative to sample size. De-
55 cisions to adjust must not depend on observed baseline balance, a principle established by².
56 Across the simulation grid, ANCOVA dominates the unadjusted analysis on efficiency without
57 compromising coverage, while the test-then-adjust strategy shows systematic over-coverage
58 relative to nominal 95%.

59 **Conclusions.** The objective criteria provide an applied investigator with a procedural route
60 from trial design to analysis-model specification that aligns with the existing theoretical
61 and regulatory consensus. The criteria are not new findings but are a consolidation of dis-
62 persed guidance into practitioner-usable form, with explicit citation of the underlying primary
63 sources; this paper is a tutorial/review, not a report of novel methodology.

64 2 Introduction

65 Randomized clinical trials (RCTs) constitute the most rigorous method for evaluating treat-
66 ment effects in medicine. A fundamental analytical decision in every RCT is whether and how
67 to adjust for baseline covariates in the primary analysis. Despite decades of statistical research

68 and increasingly clear regulatory guidance, confusion persists. Investigators routinely test for
69 baseline balance between treatment groups (and adjust only if imbalances are detected), omit
70 covariates from the primary analysis to preserve ‘simplicity,’ or select covariates post hoc
71 based on observed associations with the outcome. Each of these practices is suboptimal or
72 outright harmful. We review the theoretical, empirical, and regulatory evidence bearing on
73 covariate adjustment and synthesize it into a set of objective criteria for when and how to
74 adjust.

75 2.1 The Problem of Baseline Imbalance

76 Randomization guarantees that, on average across repetitions, baseline covariates will be
77 balanced between treatment groups. It does not guarantee balance in any individual trial.³
78 demonstrated that even non-significant imbalances in prognostic factors can strongly influence
79 the observed treatment effect, rendering conclusions misleading. The practice of testing for
80 baseline balance and adjusting only when a difference is ‘significant’ was criticized sharply by²,
81 who argued it is ‘philosophically unsound, of no practical value and potentially misleading.’
82 Senn’s position is that significance tests for baseline balance merely test whether randomiza-
83 tion was conducted properly, not whether observed imbalances affect the trial result.

84 Surveys of published trials confirm that these problematic practices remain widespread.⁴ ex-
85 amined 50 trial reports in four major journals, finding that most trials emphasized simple
86 unadjusted results, that covariate adjustment usually made negligible difference, and that
87 subgroup analyses were routinely overinterpreted. A companion paper by the same group⁵
88 highlighted the underuse of appropriate interaction tests and the inconsistency with which
89 covariates were selected and adjusted for across studies.⁶ showed that among trials using
90 stratified randomization, only 14 of 41 eligible studies adjusted for the stratification variables,
91 leading to overly conservative inference with inflated confidence intervals and reduced power.

92 2.2 Theoretical Foundations

93 2.2.1 Analysis of Covariance in the Linear Setting

94 The classical statistical argument for covariate adjustment rests on the analysis of covariance
95 (ANCOVA). When the outcome Y depends on a baseline covariate X and treatment Z , the
96 ANCOVA model

$$97 Y_i = \beta_0 + \beta_1 Z_i + \beta_2 X_i + \epsilon_i$$

98 yields a more precise estimate of β_1 than the unadjusted comparison of means, provided X is
99 prognostic for Y . The variance reduction is proportional to $R_{Y|X}^2$, the proportion of outcome
100 variance explained by the covariate. This result holds regardless of whether randomization
101 produced balanced or imbalanced groups.^{7,8} showed that ANCOVA is superior to the analysis
102 of change scores (the ‘simple analysis of change scores,’ or SACS) and that ANCOVA does not
103 require baseline balance, even in expectation, to provide unbiased treatment effect estimates.
104 This resolves Lord’s paradox in the clinical trial setting⁸.

105 ⁹ provided a concise tutorial confirming that ANCOVA is the preferred analysis whenever a
106 baseline measurement of the outcome exists, yielding estimates with smaller standard errors

107 and greater statistical power than either the comparison of follow-up values alone or the
108 analysis of change scores.

109 ¹⁰ formally studied the relative efficiency of the change-score estimator, the follow-up-only
110 estimator, and several ANCOVA variants in the context of a pretest-posttest trial. They
111 showed that ANCOVA with a treatment-by-covariate interaction term (ANCOVA II) and
112 the GEE approach are most efficient, while even the simpler ANCOVA without interaction
113 dominates both the change-score and follow-up-only estimators for all values of the baseline-
114 outcome correlation.

115 **2.2.2 The Freedman-Lin Debate**

116 ¹¹ challenged the use of OLS regression adjustment in randomized experiments under Ney-
117 man’s randomization inference framework. He showed that adjustment can worsen asymptotic
118 precision, that standard errors from the OLS model are not generally valid, and that small-
119 sample bias exists. These results generated considerable concern.

120 ¹² reexamined Freedman’s critique and showed that the essential problem was the omission
121 of treatment-by-covariate interactions. When a full set of treatment-covariate interactions
122 is included in the regression, OLS adjustment cannot hurt asymptotic precision, and the
123 Huber-White sandwich variance estimator provides valid standard errors. In balanced two-
124 group designs, even the simpler model without interactions is asymptotically valid because,
125 as Lin put it, ‘two wrongs make a right’ – the overadjustment for one group cancels the
126 underadjustment for the other.

127 **2.2.3 Semiparametric Efficiency Theory**

128 The most rigorous theoretical framework for covariate adjustment in RCTs comes from semi-
129 parametric efficiency theory.¹³ and¹⁴ developed a class of semiparametric estimators for the
130 average treatment effect that are guaranteed to be at least as efficient as the unadjusted
131 estimator and typically much more so. The key insight is that in a randomized trial, any
132 function of the baseline covariates can be incorporated into the estimator to reduce variance
133 without introducing bias, provided the estimator is constructed properly using augmented
134 inverse probability weighting or equivalent approaches.

135 ¹⁵ proved that for a large class of commonly used regression models – including linear and
136 logistic regression – standard hypothesis tests using robust (sandwich) variance estimators
137 are guaranteed to have correct Type I error asymptotically, even when the regression model
138 is misspecified. This provides theoretical justification for the common practice of fitting a
139 potentially misspecified working model and using robust standard errors.

140 ¹⁶ extended these ideas to binary outcomes using targeted maximum likelihood estimation
141 (TMLE), showing that the marginal risk difference can be estimated efficiently by fitting a lo-
142 gistic regression model and then marginalizing (standardizing) over the covariate distribution.
143 This approach avoids the well-known problem identified by¹⁷ that covariate adjustment in lo-
144 gistic regression can reduce precision of the conditional odds ratio. As¹⁶ and¹³ later clarified,
145 this problem does not arise when the target is the marginal effect.

146 **2.2.4 Survival Outcomes**

147 Covariate adjustment in the survival analysis setting has its own theoretical development.¹⁸
148 showed how auxiliary covariates can be incorporated into the log-rank test to improve
149 efficiency.¹⁹ proposed a nonparametric covariate-adjusted log-rank test with a simple closed
150 form that achieves guaranteed efficiency gains over the unadjusted test and applies univer-
151 sally across simple and covariate-adaptive randomization schemes.²⁰ developed improved
152 estimators for the restricted mean survival time that leverage baseline covariates without
153 assuming proportional hazards.

154 **2.3 Regulatory Guidance**

155 **2.3.1 ICH E9 (1999) and E9(R1) (2019)**

156 The ICH E9 guideline²¹ advises investigators to ‘identify covariates expected to have an impor-
157 tant influence on the primary variable’ and ‘specify how to account for these in the analysis,
158 in order to improve the precision of the treatment comparisons.’ The E9(R1) addendum²²
159 introduced the estimand framework, which requires precise definition of the treatment effect
160 of interest. This framework makes the distinction between conditional and marginal effects
161 particularly salient for covariate adjustment.

162 **2.3.2 EMA Guideline (2015)**

163 The European Medicines Agency guideline²³ provides concrete recommendations: (i) the pri-
164 mary analysis should include only pre-specified covariates, (ii) no treatment-by-covariate in-
165 teraction terms should appear in the primary model, (iii) stratification variables used in the
166 randomization should generally be included as covariates, and (iv) covariates should be mea-
167 sured before randomization.

168 **2.3.3 FDA Guidance (2023)**

169 The FDA’s final guidance on covariate adjustment²⁴ represents the most comprehensive reg-
170 ulatory position to date. Key recommendations include:

- 171 • Pre-specification of covariates in the statistical analysis plan
- 172 • Use of ANCOVA for continuous outcomes
- 173 • For nonlinear models (logistic, Cox), use of standardization (g-computation) to estimate
174 marginal effects
- 175 • Use of robust (sandwich) or bootstrap standard errors
- 176 • Inclusion of stratification factors as covariates
- 177 • Avoidance of data-driven covariate selection

178 The FDA guidance was informed directly by the semiparametric theory of¹³ and the practical
179 work of^{25, 26}, and others.

2.4 The Conditional versus Marginal Distinction

A critical conceptual issue for nonlinear models is the distinction between conditional and marginal treatment effects. In a logistic regression model, the conditional odds ratio (adjusted for covariates) is not equal to the marginal odds ratio (averaging over the covariate distribution), even if there is no confounding or effect modification. This ‘noncollapsibility’ has generated confusion about whether covariate adjustment ‘changes the estimand.’

²⁷ addressed this directly, arguing that if the target parameter is carefully defined, adjustments can be used with minimal assumptions and the benefit of increased efficiency is achieved at essentially no cost.²⁸ reviewed the available methods and advocated for the standardization approach, which leverages the covariate-outcome relationship for efficiency while targeting a well-defined marginal effect.²⁹, in a commentary on this work, agreed that covariate adjustment ‘accounts for outcome heterogeneity and prevents power loss,’ but cautioned against the marginal/standardization framing, arguing that conditional models preserve the full information content of the covariates and that marginal effects can obscure treatment effect heterogeneity.

2.5 Seven Myths and Persistent Misconceptions

³⁰ enumerated seven common misconceptions about randomization, several of which bear directly on covariate adjustment:

1. *Balance is necessary for valid inference.* False: randomization justifies inference regardless of balance; adjustment is for efficiency, not validity.
2. *Observed covariates may be ignored because one has randomized.* False: ignoring prognostic covariates wastes information and reduces power.
3. *Randomization is inefficient.* False: combined with appropriate analysis, randomization plus covariate adjustment is highly efficient.

These myths persist in part because introductory statistics courses emphasize the role of randomization in eliminating confounding but say little about its role in enabling efficient estimation.

2.6 Empirical Evidence on Efficiency Gains

Multiple empirical studies have quantified the practical benefits of covariate adjustment.³¹ showed via simulation that pre-specified adjustment for prognostic covariates in trials with binary outcomes can reduce required sample sizes by 3% to 46%, depending on the strength of the covariate-outcome association.³² demonstrated a 25% reduction in sample size requirements in traumatic brain injury (TBI) trials when adjusting for seven baseline predictors in the IMPACT dataset.³³ reanalyzed the CRASH trial (n = 9497), finding that covariate adjustment using validated prognostic models substantially increased power.³⁴ showed that the difference between adjusted and unadjusted effects can be substantial in the presence of outcome heterogeneity, with adjustment providing a more accurate estimate of the treatment benefit.

218 ²⁶ demonstrated efficiency gains equivalent to a 22-30% reduction in sample size in
219 simulations.²⁵ applied semiparametric methods to COVID-19 treatment trials, achieving
220 meaningful precision gains for binary, ordinal, and time-to-event outcomes.³⁵ proposed a
221 method for adjusting via a prognostic score trained on external (historical) data, showing
222 that the approach attains the semiparametric efficiency bound under certain conditions.

223 2.7 Practical Guidance for Covariate Selection

224 ⁷ provided the first systematic framework for selecting covariates. Their key recommendations
225 are: (i) covariates balanced by design (stratification factors) should always be included, (ii)
226 strongly prognostic covariates should be included regardless of balance, (iii) data-dependent
227 selection methods based on the current trial data should be avoided as they produce mis-
228 leading inference, and (iv) all decisions should be pre-specified in the protocol based on prior
229 knowledge.

230 ³⁶ offered a comprehensive practical guide for planning covariate adjustment in individually
231 randomized trials. They compared three approaches – direct adjustment, standardization,
232 and inverse probability of treatment weighting – concluding that all three are asymptotically
233 efficient and similar in large samples, but that standardization should be preferred when
234 the target is a marginal estimand because it avoids convergence problems with nonstandard
235 models.

236 ³⁷ addressed the question of how many covariates to include, showing that the benefits of
237 adjustment diminish and can reverse when too many weakly prognostic covariates are added
238 relative to the sample size.

239 ³⁸ examined adjustment in nonlinear models, confirming that the standard advice for linear
240 models does not always transfer and that care is needed with the choice of estimand and
241 variance estimation.

242 ³⁹ provided a synthesis linking covariate-adaptive randomization, stratification, and covariate-
243 adjusted analysis, with practical recommendations for modern trial design and analysis.

244 2.8 Present Study

245 The preceding literature provides a strong theoretical and empirical foundation, and several
246 sources already give practitioner-facing recommendations that overlap substantially with what
247 follows, including the FDA and EMA guidance documents themselves,³⁶'s practical guide to
248 planning adjustment,⁴⁰'s empirical assessment of the consequences of (not) adjusting, and²⁸'s
249 review of conditional-versus-marginal considerations. This paper does not claim a new empiri-
250 cal finding or a novel estimator; its contribution is a single, citation-backed decision procedure
251 that maps each of these dispersed sources onto three operational questions – when to adjust,
252 which covariates to adjust for, and how to specify the adjustment – together with a worked
253 illustration of why the common test-then-adjust heuristic fails. We synthesize the evidence
254 into these decision rules and illustrate the efficiency and coverage consequences through a
255 small Monte Carlo simulation.

256 3 Methods

257 3.1 Simulation Design

258 We conduct a Monte Carlo simulation, reported following the ADEMP framework of¹ (Aims,
259 Data-generating mechanisms, Estimands, Methods, Performance measures), to illustrate the
260 efficiency gains from ANCOVA relative to the unadjusted comparison of means, across a range
261 of covariate-outcome correlations. The simulation also illustrates that testing for baseline
262 balance and adjusting only when a significant imbalance is detected (the ‘test-then-adjust’
263 strategy) does not improve – and can distort the coverage of – inferential properties. The
264 simulation varies the covariate-outcome correlation (γ) and sample size (n) only; it does not
265 vary baseline imbalance as a design factor, and it is restricted to a single continuous outcome
266 and a correctly specified adjustment model (see Limitations).

267 3.2 Data Generating Model

268 For each replication $b = 1, \dots, B$:

- 269 1. Generate n subjects, randomized 1:1 to treatment ($Z = 1$) or control ($Z = 0$).
- 270 2. Generate a baseline covariate $X_i \sim \text{Normal}(0, 1)$.
- 271 3. Generate the outcome as

$$272 \quad Y_i = \beta_0 + \tau Z_i + \gamma X_i + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

273 where $\tau = 0.3$ is the true treatment effect, γ controls the strength of the prognostic
274 covariate, and $\sigma^2 = 1 - \gamma^2$ so that $\text{Var}(Y) = 1$ marginally.

275 3.3 Analysis Strategies

276 For each simulated dataset, we apply three analysis strategies:

- 277 1. **Unadjusted:** Two-sample t -test comparing treatment group means.
- 278 2. **Always-adjust (ANCOVA):** Linear regression of Y on Z and X , with robust (HC2)
279 standard errors.
- 280 3. **Test-then-adjust:** First test $H_0 : \bar{X}_1 = \bar{X}_0$ at $\alpha = 0.10$. If rejected, use ANCOVA;
281 otherwise use the unadjusted analysis.

282 3.4 Parameter Values

Table 1: Simulation parameters.

Parameter	Values
Sample size (n)	100, 200
Treatment effect (τ)	0.3
Covariate effect (γ)	0, 0.3, 0.5, 0.7
Residual variance (σ^2)	$1 - \gamma^2$
Number of replications (B)	5000

3.5 Metrics

For each scenario and analysis strategy, we compute:

- **Bias:** Mean $\hat{\tau} - \tau$
- **Empirical SE:** Standard deviation of $\hat{\tau}$ across replications
- **Mean model SE:** Mean of estimated standard errors across replications
- **Coverage:** Proportion of 95% CIs containing τ
- **Power:** Proportion of replications rejecting $H_0 : \tau = 0$ at $\alpha = 0.05$ (two-sided)

4 Results

4.1 Efficiency Gains from ANCOVA

Table 2: Simulation results across analysis strategies and covariate strengths ($B = 5000$ replications per scenario). MCSE columns are Monte Carlo standard errors for coverage and power, following the ADEMP framework of Morris, White, and Crowther (2019).

n	γ	Strategy	Bias	Emp. SE	Mean SE	Coverage	MCSE(Cov.)	Power	MCSE(Pow.)
100	0.0	Unadjusted	-0.002	0.197	0.199	0.952	0.003	0.310	0.007
100	0.0	ANCOVA	-0.002	0.199	0.200	0.952	0.003	0.307	0.007
100	0.0	Test-then-adjust	-0.002	0.198	0.200	0.951	0.003	0.310	0.007
200	0.0	Unadjusted	0.000	0.139	0.141	0.955	0.003	0.562	0.007
200	0.0	ANCOVA	0.000	0.140	0.142	0.956	0.003	0.557	0.007
200	0.0	Test-then-adjust	0.000	0.140	0.141	0.956	0.003	0.560	0.007
100	0.3	Unadjusted	0.004	0.199	0.199	0.951	0.003	0.328	0.007
100	0.3	ANCOVA	0.003	0.191	0.191	0.951	0.003	0.351	0.007
100	0.3	Test-then-adjust	0.003	0.196	0.199	0.953	0.003	0.324	0.007
200	0.3	Unadjusted	0.000	0.139	0.141	0.954	0.003	0.568	0.007
200	0.3	ANCOVA	0.000	0.134	0.135	0.951	0.003	0.600	0.007
200	0.3	Test-then-adjust	0.000	0.136	0.141	0.956	0.003	0.570	0.007
100	0.5	Unadjusted	0.001	0.201	0.200	0.953	0.003	0.322	0.007
100	0.5	ANCOVA	0.001	0.176	0.174	0.947	0.003	0.404	0.007
100	0.5	Test-then-adjust	0.003	0.190	0.198	0.961	0.003	0.327	0.007
200	0.5	Unadjusted	0.002	0.143	0.141	0.947	0.003	0.566	0.007
200	0.5	ANCOVA	0.002	0.124	0.123	0.945	0.003	0.681	0.007
200	0.5	Test-then-adjust	0.002	0.136	0.139	0.957	0.003	0.579	0.007
100	0.7	Unadjusted	0.000	0.202	0.200	0.946	0.003	0.316	0.007
100	0.7	ANCOVA	0.000	0.145	0.143	0.947	0.003	0.551	0.007
100	0.7	Test-then-adjust	-0.001	0.177	0.195	0.969	0.002	0.322	0.007
200	0.7	Unadjusted	0.003	0.142	0.141	0.949	0.003	0.569	0.007
200	0.7	ANCOVA	0.000	0.102	0.101	0.948	0.003	0.837	0.005
200	0.7	Test-then-adjust	0.002	0.125	0.137	0.971	0.002	0.600	0.007

Table 2 presents the simulation results (all figures computed directly from the saved simulation output; none are hard-coded). We observe that when the covariate is nonprognostic ($\gamma = 0$), all three strategies perform similarly, as expected. As γ increases, ANCOVA achieves progressively smaller empirical standard errors and correspondingly higher power. For $\gamma = 0.7$ (i.e., the covariate explains 49% of outcome variance), ANCOVA increases power from 0.32 to 0.55 at $n = 100$, and from 0.57 to 0.84 at $n = 200$.

298 The test-then-adjust strategy's power (0.32 at $n = 100$, $\gamma = 0.7$) is close to the unadjusted
 299 power and far below ANCOVA's, so it is more accurate to say test-then-adjust behaves like the
 300 unadjusted analysis on power rather than as an intermediate strategy. Its coverage, however, is
 301 systematically above nominal: at $n = 100$, $\gamma = 0.7$, coverage is 0.969 (Monte Carlo SE 0.002),
 302 and at $n = 200$ it is 0.971, both several Monte Carlo standard errors above 0.95. This is not
 303 model selection bias in the usual sense; it reflects that most replications are approximately
 304 balanced, so test-then-adjust falls back to the unadjusted standard error (mean model SE
 305 0.200 at $n = 100$, $\gamma = 0.7$), which is conservative relative to the empirical variability that
 306 would obtain if ANCOVA were always used (0.177 empirical SE for test-then-adjust versus
 307 0.195 mean model SE in the same scenario). The strategy is therefore strictly dominated by
 308 always-adjust: it sacrifices power relative to ANCOVA without offering a validity advantage,
 309 since ANCOVA's coverage is close to nominal in every scenario studied.

310 4.2 Relative Efficiency

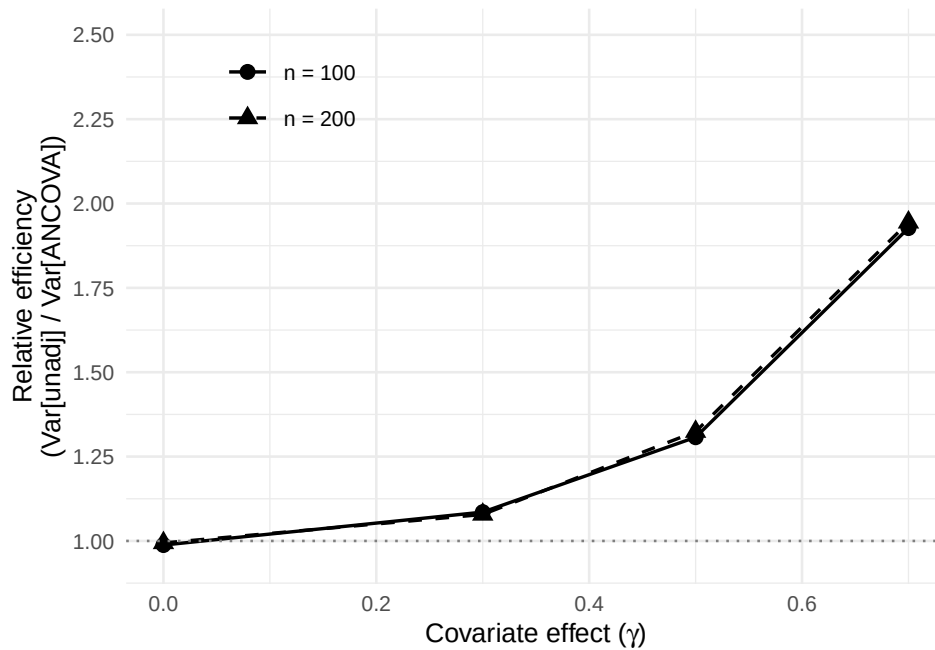


Figure 1: Relative efficiency of ANCOVA versus the unadjusted estimator as a function of covariate prognostic strength. Relative efficiency is defined as the ratio of empirical variances (unadjusted / ANCOVA).

311 We see in Figure 1 that the relative efficiency of ANCOVA increases approximately as $1/(1 -$
 312 $\gamma^2)$, consistent with theory. This ratio is asymptotically independent of sample size; the close
 313 agreement between the $n = 100$ and $n = 200$ curves here is consistent with that asymptotic
 314 result but is based on only two sample sizes and should not be read as a general empirical
 315 finding across arbitrary n .

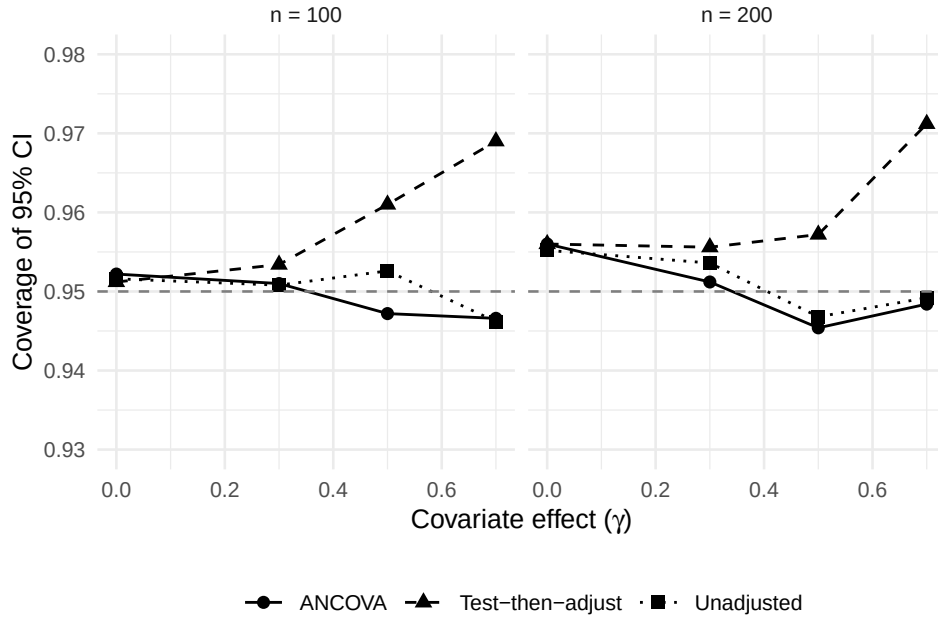


Figure 2: Empirical coverage of 95% confidence intervals across analysis strategies. The horizontal dashed line marks nominal 95% coverage.

4.3 Coverage Properties

Figure 2 shows that the unadjusted and ANCOVA strategies maintain approximately nominal coverage throughout. The test-then-adjust strategy shows systematic over-coverage that grows with γ , as quantified in Table 2 and discussed above; this is a consequence of the SE mixture described above, not a form of model-selection bias in the estimate itself (the point estimates remain approximately unbiased in every scenario, per Table 2).

5 Discussion

5.1 Summary of Objective Criteria

Based on the theoretical literature, empirical evidence, and regulatory guidance, we propose the following objective criteria for covariate adjustment in the primary analysis of an RCT.

When to adjust:

1. *Always*, if prognostic covariates are available and pre-specified. The decision to adjust should never be based on observed baseline balance^{2,30}.
2. *Always* include stratification variables used in the randomization design as covariates in the analysis^{6,23,24}.
3. *Always* include the baseline measurement of the outcome variable, if available⁸⁻¹⁰.

Which covariates to adjust for:

4. Select covariates based on prior knowledge of their prognostic value – from previous trials, meta-analyses, or established clinical understanding^{7,37}.

- 335 5. The number of covariates should be small relative to the sample size.⁷ provide a formula
336 relating the benefit of adjustment to the number of subjects per covariate.
337 6. Do not select covariates based on the trial data itself (neither on baseline imbalance
338 nor on observed covariate-outcome associations). Data-driven selection produces biased
339 inference^{4,7}.

340 **How to adjust:**

- 341 7. For continuous outcomes: use ANCOVA (linear regression of Y on Z and X). Standard
342 (OLS) or robust standard errors are both acceptable for balanced designs^{12,24}.
343 8. For binary outcomes: fit a logistic regression model and compute the marginal effect
344 (standardization / g-computation) to obtain a risk difference or relative risk^{16,28}. Al-
345 ternatively, use a linear probability model or modified Poisson regression with robust
346 standard errors.
347 9. For time-to-event outcomes: use a covariate-adjusted log-rank test^{18,19} or estimate the
348 restricted mean survival time with covariate adjustment²⁰.
349 10. Use robust (sandwich) standard errors to guard against model misspecification^{13,15,24}.
350 11. Pre-specify all covariates and the analysis method in the statistical analysis plan. The
351 primary analysis should contain no interaction terms²³.

352 **5.2 Consistency with Regulatory Guidance**

353 These criteria are fully consistent with the EMA guideline²³ and the FDA final guidance²⁴.
354 Both agencies endorse pre-specified covariate adjustment as a tool for improving precision
355 and recommend against data-driven covariate selection. The FDA guidance specifically rec-
356 ommends standardization for nonlinear models and robust variance estimation.

357 **5.3 Limitations**

358 The simulation presented here is deliberately simple and illustrative, not a comprehensive eval-
359 uation of the eleven criteria. It is restricted to a single continuous outcome, a single covariate,
360 a correctly specified linear data-generating model, and no missing data. It does not vary
361 baseline imbalance as a design factor and does not include a binary or time-to-event scenario,
362 so criteria 8-10 (nonlinear and survival outcomes) rest on the cited theoretical and applied
363 literature rather than on evidence generated here. Real trials involve multiple covariates, var-
364 ious outcome types, missing data, and complications such as noncompliance and intercurrent
365 events. The estimand framework of ICH E9(R1)²² introduces additional considerations about
366 what exactly is being estimated. The core qualitative message – that pre-specified covariate
367 adjustment improves efficiency without sacrificing validity – is supported by the broader liter-
368 ature cited throughout this paper across outcome types and regulatory frameworks, but the
369 simulation itself demonstrates this only for the continuous-outcome, correctly specified case.

370 The simulation uses heteroskedasticity-consistent (HC2) standard errors for the ANCOVA
371 and test-then-adjust strategies, computed via `sandwich::vcovHC()`, matching the Methods
372 description and the FDA’s recommendation for robust variance estimation. For the balanced,
373 correctly-specified designs considered here HC2 and classical OLS standard errors are asymp-

374 totically equivalent¹²; the sandwich estimator was retained because it remains valid under
375 the misspecification and heteroscedasticity that can arise in practice, even though neither is
376 present by construction in this simulation.

377 6 Future Research

378 Several directions for future investigation emerge from this review:

- 379 1. **Machine learning for covariate adjustment.** The FDA guidance does not yet
380 address the use of machine learning methods for constructing prognostic scores.³⁵ showed
381 that a prognostic score trained on external data can achieve the efficiency bound, but the
382 operating characteristics of cross-validated or penalized approaches in the trial data itself
383 remain under study.⁴¹ explored combining covariate adjustment with group sequential
384 and information-adaptive designs.
- 385 2. **Missing covariate data.** When baseline covariates are missing, the analyst faces a
386 tradeoff between the efficiency gains from adjustment and the potential bias from ad
387 hoc handling of missingness. Multiple imputation and augmented estimators are active
388 research areas.
- 389 3. **Cluster-randomized and crossover trials.** The theory reviewed here applies pri-
390 marily to individually randomized, parallel-group trials. Extensions to cluster random-
391 ization and crossover designs present additional methodological challenges.
- 392 4. **Software tools.** The R package `estimatr` implements Lin’s regression estimator via
393 the `lm_lin()` function. Broader software implementing the standardization approach
394 for various outcome types would make best practices more accessible.
- 395 5. **Harmonization of conditional and marginal frameworks.** The tension be-
396 tween conditional effects (natural for clinicians reasoning about individual patients)
397 and marginal effects (natural for regulators reasoning about populations) remains
398 unresolved^{27–29}.

399 7 Conflict of interest

400 The author declares no conflicts of interest.

401 8 Funding

402 This work received no specific funding.

403 9 Data availability statement

404 The simulation function, the driver script that runs it, the saved simulation output, and the
405 manuscript source are all in the `baselinedist` project repository. Specific paths are listed in
406 the Reproducibility section below.

407 10 Reproducibility

408 This manuscript was rendered with R 4.4.x. The simulation function `run_sim()` lives
409 in `R/run_sim.R` as a documented, exported package function and is unit-tested in
410 `inst/tinytest/test_run_sim.R`. It uses base R `lm()` for model fitting, `sandwich::vcovHC()`
411 (HC2) for the ANCOVA and test-then-adjust standard errors, `tibble` for the returned sum-
412 mary, `dplyr` and `ggplot2` for reporting, and `rmarkdown` plus `bookdown` for the manuscript
413 build. MASS is not used; the single covariate is generated with `stats::rnorm()`.

414 **Random-number generator.** `analysis/scripts/run_simulation.R` calls `run_sim(...,`
415 `seed = master_seed + scenario_id)` once per scenario, with `master_seed <- 20260305`,
416 so each of the eight (n, γ) scenarios has its own reproducible seed set at the top of `run_sim()`.
417 There is no per-replicate reseeding within a scenario; replicates within a scenario share one
418 continuous RNG stream started from that scenario's seed.

419 **Data and code availability.** The simulation is not embedded in this Rmd. `analysis/scripts/run_simulation.R`
420 runs the full factorial design ($n \in \{100, 200\}$, $\gamma \in \{0, 0.3, 0.5, 0.7\}$, $B = 5000$) and
421 saves the combined results, together with the master seed and a timestamp, to
422 `analysis/data/derived_data/sim_results.rds`. This document's `simulation` chunk
423 reads that `.rds` file directly and stops with an informative error if it is missing, so the
424 manuscript can never silently render with stale or absent results. To regenerate the
425 results, run `Rscript analysis/scripts/run_simulation.R` from the repository root
426 (approximately one to two minutes on a laptop); then re-render this document with `bash`
427 `tools/render.sh analysis/report/report.Rmd`. The manuscript source, bibliography
428 (`references.bib`), and CSL file are at `analysis/report/`.

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